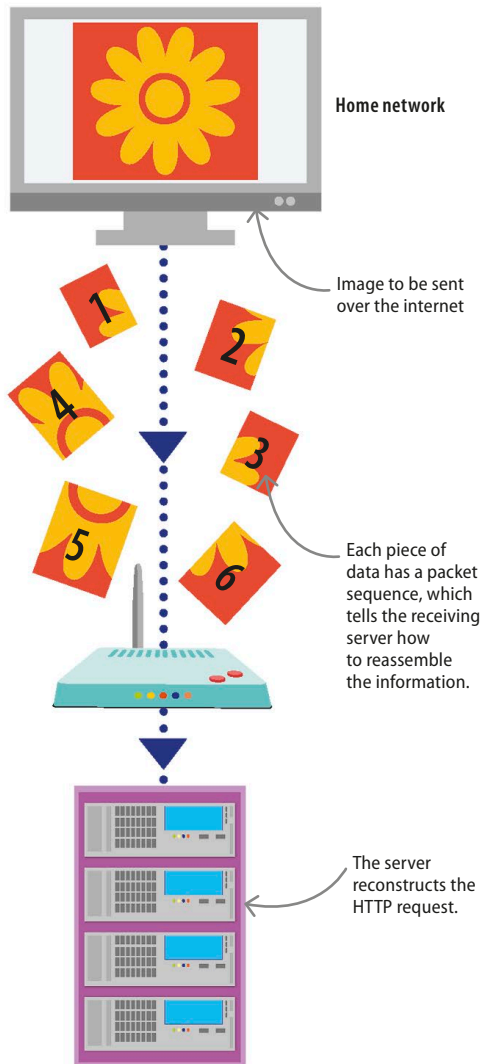


Packets

Transmitting large files all at once clogs up networks and prevents other computers from sending or receiving messages. To solve this issue, most protocols break data up into small packets. Pictures, text messages, and even basic HTTP requests are sliced up and transferred piece by piece. Each packet is given a header that contains its destination's IP address and a return IP address.

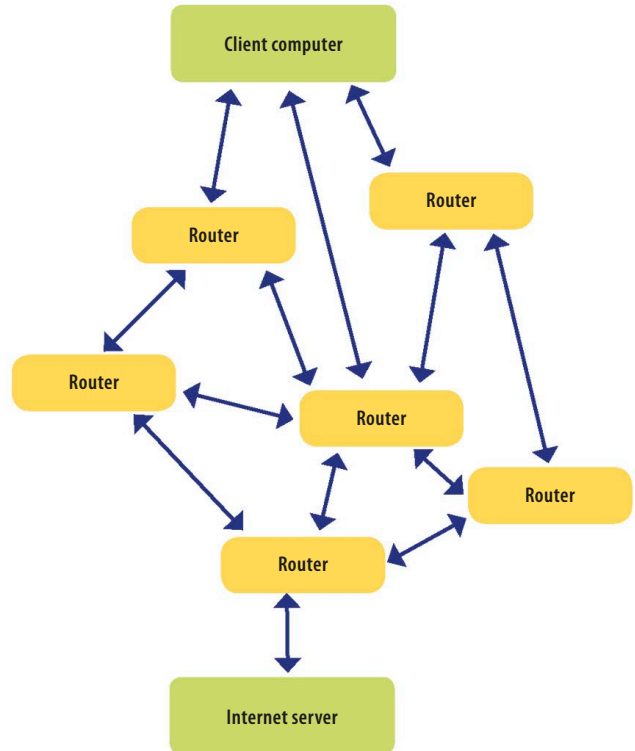


△ Transmission Control Protocol (TCP)/ Internet Protocol (IP)

In TCP/IP, each packet is assigned a number to help the target computer reassemble the packets into the original message.

Routing

Transmitting a message works a bit like a relay race. At each hop, the current router checks the message's target IP and determines the best router to send the message to next. This continues until the destination is reached. It's therefore possible that packets from the same message end up taking different routes to the same computer.



△ Carrier-Sense Multiple Access (CSMA)

CSMA is a low-level protocol that handles the process of transmitting packets through a wire. It includes checking that the route is clear, and helps avoid collisions.

IN DEPTH

Security and encryption

Since data packets often take convoluted routes through strange routers, there are many ways for hackers to intercept messages. Encryption is a simple way to protect data. The basic idea is to use a secret key to transform a message into gibberish. Only authorized contacts who have a matching key can recover the message. That way, even if hackers manage to steal all of the packets, the information is useless. Several protocols, such as HTTPS, have built-in encryption.